Traffic Sign Classification on GTSRB

Accuracy vs. Adversarial Robustness

-- Comparing MLP, CNN, and AugMix CNN under I-FGSM attacks

XIAOHAN XUE 313000

TL;DR

Motivation:

- Autonomous driving systems need reliable traffic sign recognition because signs affect driving decisions such as speed limits, warnings, and restrictions. In real-world settings, signs may appear under different lighting, angles, weather conditions, occlusion, or image noise.

Research problem:

- High clean-image accuracy may not guarantee robustness when traffic sign images are slightly perturbed. This project studies whether models that perform well on clean GTSRB images also remain reliable under adversarial perturbations.

Approach:

- I compare MLP, CNN, and AugMix CNN on clean test accuracy, then evaluate each model under I-FGSM attacks at multiple ε values.

Main finding:

- CNN achieves the highest clean accuracy, but MLP retains higher accuracy under I-FGSM. AugMix slightly improves CNN robustness, but the improvement is limited.

Takeaway:

- Natural variation / augmentation robustness ≠ gradient-based adversarial robustness.

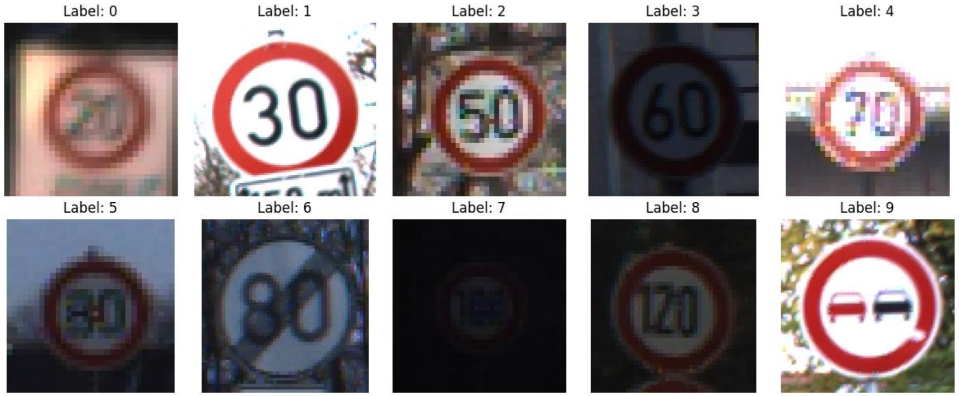
Literature Review

Prior work / method	Role in this project
GTSRB benchmark	Provides the standard traffic sign classification task with 43 classes.
CNN-based traffic sign recognition	Shows why convolutional architectures are strong for GTSRB and motivates my CNN model.
FGSM / I-FGSM attacks	Provides the adversarial evaluation method used to test model sensitivity to small perturbations.
AugMix	Provides the augmentation and JSD consistency-loss idea used to test whether CNN robustness can improve.
Efficient Traffic Sign Recognition paper	Shows prior work can achieve very high clean GTSRB accuracy, motivating my focus on robustness beyond clean accuracy.

Data Preprocessing

Dataset — German Traffic Sign Recognition Benchmark

- 43 classes, ~39 K training images, ~12.6 K test images
- Images are in different size, lighting, clearness, occlusion, and viewing angle
- Class imbalance exists across traffic sign categories
    - E.g. label 1,2 have 1500 examples each, while label 0,19,37 only have 150 examples each



- Resize all images to 32 x 32 px and convert to RGB
- Convert images to tensors, scaling pixel values to [0,1], then standardize using training-set mean and standard deviation
- Apply augmentation to test whether robustness can be improved

Model Design: Testing Accuracy vs. Robustness

MLP

- 3 hidden fully connected layers
- batchNorm + Dropout for stable training and regularization
- Baseline model - Tests model capacity without spatial inductive bias

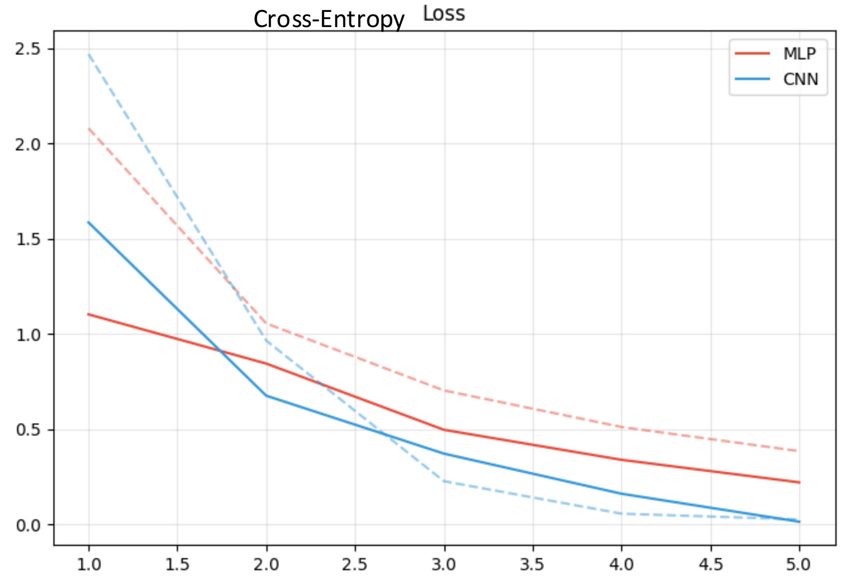
CNN

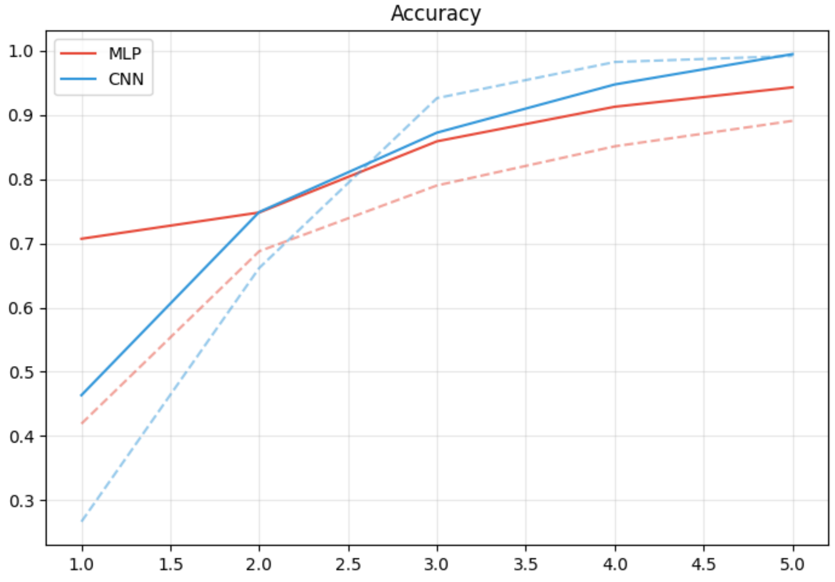
- Compact VGG-style sequential CNN
- Three convolutional blocks; each block has two conv layers before max pooling
- Tests whether local receptive fields and weight sharing improve clean accuracy

AugMix CNN

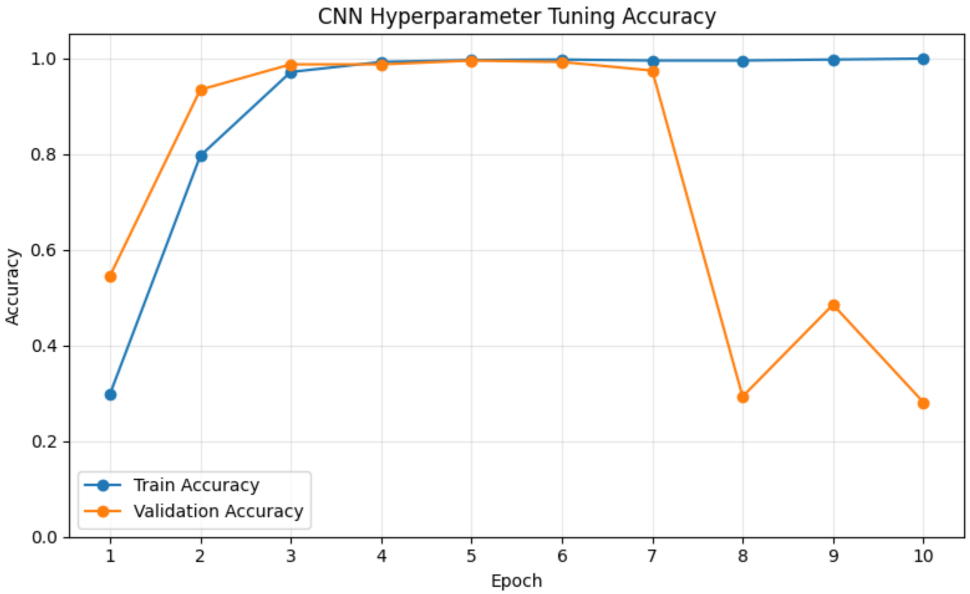
- Trains on the original image plus two randomly augmented versions per step
- Uses JSD consistency loss to reduce prediction changes across views
- Tests whether augmentation improves robustness

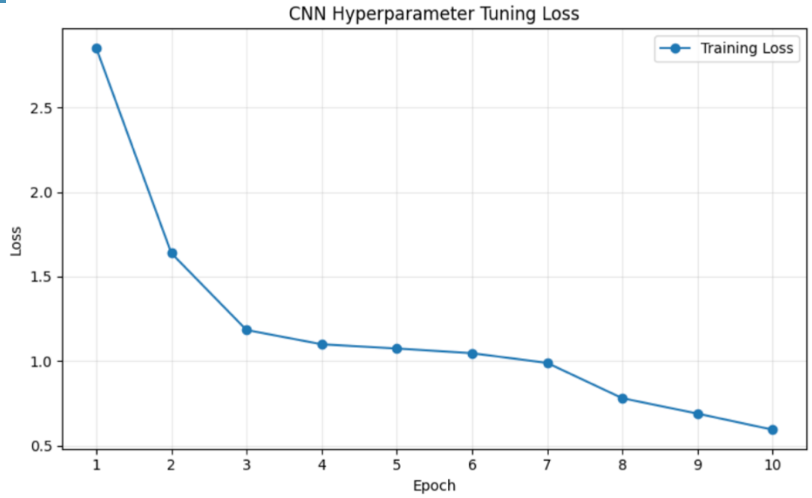
Learning curve Comparison





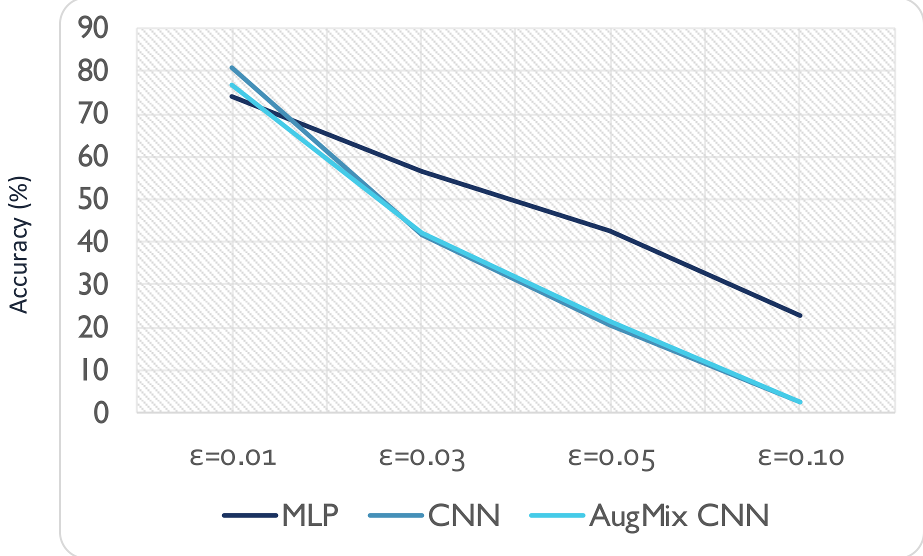
- Dashed – training set, solid – validation set
- CNN starts higher in entropy loss but converges much faster and lower than MLP; MLP is more stable from the start but ends up with higher loss.
- MLP starts higher in accuracy (epoch 1 already ~70%) but CNN catches up and stays higher than MLP





- CNN validation accuracy increases quickly and peaks around epoch 5
- After epoch 7, validation accuracy drops sharply while training accuracy remains high
- This suggests overfitting or training instability after the best checkpoint
- The best validation checkpoint is used for final clean and adversarial evaluation

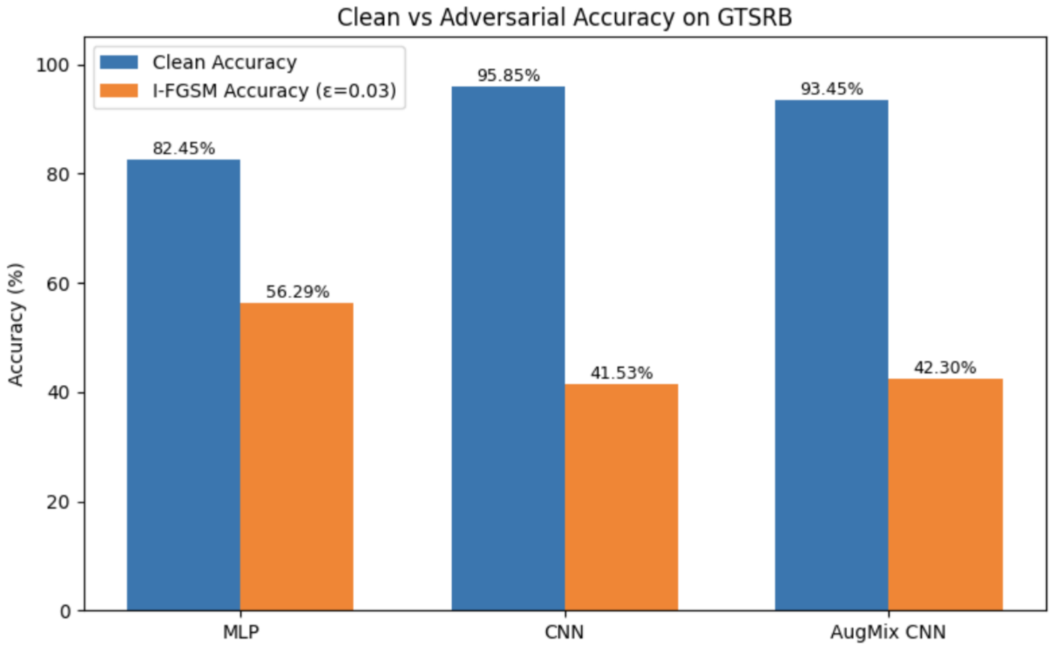
Adversarial Robustness: Does Clean Accuracy Transfer?



- I-FGSM adds small repeated gradient-based perturbations to test model sensitivity
- CNN has the highest clean accuracy, but its adversarial accuracy drops sharply as ε increases
- MLP has lower clean accuracy, but retains higher accuracy at stronger perturbations
- AugMix slightly improves CNN robustness, but the improvement is limited

Model	Clean Accuracy	ε=0.01	ε=0.03	ε=0.05	ε=0.10
MLP	82.45%	73.98%	56.29%	42.34%	22.85%
CNN	95.85%	80.81%	41.53%	20.40%	2.49%
AugMix CNN	93.45%	76.76%	42.30%	21.58%	2.58%

Accuracy v.s. robustness Tradeoff



- CNN performs best on clean images.
- MLP performs worse on clean images but is less affected by I-FGSM.
- AugMix slightly improves CNN robustness, from 41.53% to 42.30%
- Higher clean accuracy does not necessarily mean higher adversarial robustness.

Summary

Research question:

For traffic sign classification in autonomous-driving settings, higher clean accuracy does NOT imply stronger robustness under input perturbations

- CNN achieved the best clean accuracy, but MLP retained higher adversarial accuracy under I-FGSM.

Key findings:

- CNN achieved the highest clean accuracy on test set: **95.85%**.
- MLP had lower clean accuracy: **82.45%**, but higher I-FGSM accuracy at $\varepsilon = 0.03$: **56.29%**. CNN dropped to **41.53%** under the same attack.
- AugMix CNN slightly improved CNN adversarial accuracy to **42.30%**, but the improvement was limited.

Final takeaway:

In autonomous-driving applications, a model that performs well on clean images may still be fragile under small perturbations. Robustness testing is necessary before trusting a traffic sign classifier in safety-related settings.

Links & Resources

Code

Project Notebook (GitHub)

Dataset

GTSRB on HuggingFace (tanganke/gtsrb)

Original GTSRB benchmark

Key Papers

Sermanet & LeCun (2011) — CNN on GTSRB

Goodfellow et al. (2014) — FGSM

Kurakin et al. (2016) — I-FGSM / BIM

Hendrycks et al. (2020) — AugMix

Demo

YouTube — Pre-recorded demonstration

Replace bracketed placeholders with your actual GitHub, YouTube, and any personal links.